

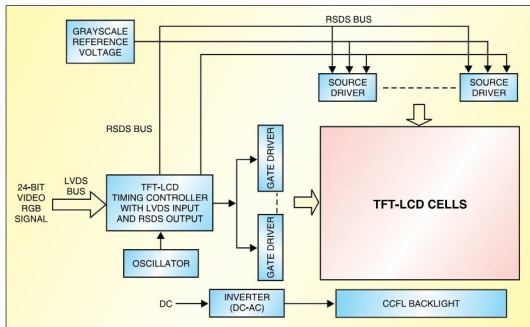


Fig. 9: Block diagram of TFT-LCD panel controller/driver

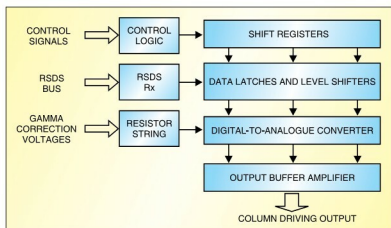


Fig. 10: Source driver architecture

the gate of TFTs in a specific row and thus allows line-by-line scanning from top to bottom.

The source driver consists of a shift register, data latch, level shifter, digital-to-analogue converter (DAC) and output buffer.

Digital data is loaded and stored in data latches until conversion takes place. Because each output has an independent DAC, data conversion is required only once per line. The DAC converts the digital gray-scale value into an equivalent analogue voltage with very fast conversion rate and this analogue voltage is fed to TFT-LCD cells through the output buffer.

The DAC also allows gamma correction. In a TFT-LCD, TFTs are

controls the overall brightness of images displayed on the screen.

Gray-scale and colour generation

The luminance of light depends on the voltage applied between the two transparent electrodes of a liquid crystal. Gray levels are an intermediate level of brightness between the brightest of bright and the darkest of dark that a unit pixel can generate by varying the voltage between the electrodes. This gray-scale voltage can be generated either in analogue or digital form. Gray-scale voltage in analogue form, i.e., continuous range, is generated using a voltage divider. The DC-DC converter can also be used to obtain

in-between voltages for gray-scale generation.

In the digital method, gray-scale voltage is defined in 8-, 10- and 12-bit forms. Hence the number of gray levels is determined by the number of bits used to represent the gray-scale voltage. For example, if eight bits are used to represent gray-scale voltage, the number of possible gray levels will be 2^8 (= 256).

The colour filter of a TFT-LCD consists of three primary colours: red, green and blue. In a colour TFT-LCD, each pixel is a combination of three subpixels having red (R), green (G) and blue (B) colours. Because the subpixels are too small to

distinguish them independently, practically any colour produced is a mixture of these primary colours. The video signal is composed of R, G and B colour signals and 'n' bits are used to represent each colour. The number of colours produced is determined by the combinations of R, G and B that are possible with the number of gray-scale levels: If an LCD displays images with 8-bit gray-scale, it will display 2^8 (=256) shades for red, green and blue colour each. Thus the total number of different colours, known as TrueColour, is $2^8 \times 2^8 \times 2^8$ (=16.78 million).

If you compare 8- and 10-bit gray-scale displays, you'll find that 10-bit gray-scale images are softer and more natural than 8-bit gray-scale images. This is because 10-bit gray-scale gives 2^{10} (=1024) shades for each primary colour and the total number of colours (natural colours) is $2^{10} \times 2^{10} \times 2^{10}$ (=1.07 billion). A 12-bit gray-scale has also been tried. It gives much more realistic picture. ●

Daniel A. Figueiredo, an M.Tech in electronics design and technology from C.E.D.T. of India, Aurangabad, is a member of IEEE, USA, and has 29 years of experience in varied roles. Harshal R. Patil is pursuing M.Tech in electronics design and technology at DOEACC Centre (Formerly C.E.D.T. of India), Aurangabad